Shri Ram Dwivedi

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EDUCATION

- **B. Tech Computer Science with AIML - Kalinga University, Raipur** | CGPA: 9.46/10 | **2023-2027**
- **Higher Secondary XII** - CBSE- Jawahar Navodaya Vidyalaya, Satna | Score: 74.8% | Science Stream with Computer Science **2022**
- **Secondary X** – CBSE - Jawahar Navodaya Vidyalaya, Satna | Score: 88.6% | **2020**

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, SQL, C.
- **Machine Learning & Deep Learning:** TensorFlow, PyTorch, Keras, Scikit-Learn, 3D U-Net, CNNs, RNNs, LSTMs, Transformers, Autoencoders, GANs, Supervised Learning, Unsupervised Learning, Transfer Learning, Model Fine-Tuning, Neural Network Architecture Design
- **Computer Vision:** OpenCV, MediaPipe, YOLOv8, ST-GNN, Pose Estimation, Facial Recognition, Object Detection, Image Segmentation, Action Recognition, Deepfake Detection, Real-Time Video Processing
- **Natural Language Processing (NLP):** NLTK, Hugging Face Transformers, TF-IDF, Text Classification, Sentiment Analysis, Named Entity Recognition (NER), Tokenization, Text Preprocessing, Word Embeddings
- **Large Language Models (LLMs) & Agentic AI:** Google Gemini API, Prompt Engineering, RAG (Retrieval-Augmented Generation), MCP (Model Context Protocol), LLM Orchestration, LLM Integration, Agentic AI Pipelines, Intent Classification, Memory Management, Autonomous AI Agents, Multi-Step Reasoning, n8n Workflow Automation
- **Backend & API Development:** FastAPI, Flask, REST APIs, OAuth2 Authentication, Session Management, Microservices Architecture, WebSockets
- **Cloud & DevOps:** Google Cloud Platform (GCP), Cloud Run, Compute Engine (VMs), Docker, CI/CD Pipelines, Git, GitHub
- **Data Science & Analytics:** Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, MySQL, Data Preprocessing, Feature Engineering, Exploratory Data Analysis (EDA), Statistical Analysis.

PROFESSIONAL EXPERIENCE

Founding AI Engineer | Ryze AI | Remote | June 2025 – July 2026 (1+ Years)

- Architected and shipped the end-to-end agentic AI pipeline for a production Ads Specialist Agent as founding engineer, integrating Google Gemini LLM with Google Ads API, Meta Ads API, and Google Analytics API to enable autonomous campaign optimization, budget allocation, and cross-platform performance reporting.
- Engineered MCP (Model Context Protocol) servers for agent tool integrations, standardizing LLM orchestration and secure tool access across advertising platform APIs in production ML systems.
- Scaled a multi-tenant Python microservices backend to 1,000+ concurrent users by building FastAPI services that orchestrate AI agent actions, RESTful API integrations, session management, and secure OAuth2 authentication.
- Owned the AI agent in production end-to-end deploying microservices on Google Cloud Platform (GCP) via Cloud Run and Compute Engine (VMs), then monitoring, debugging, and maintaining live systems to sustain high availability, low latency, and cost-efficient auto-scaling.
- Designed prompt engineering frameworks for production LLM workflows and a persistent agent memory system, enabling automated ad strategy generation, multi-step reasoning, long-term context retention, and continuous learning across 1,000+ user interactions.

AI Intern | Realfy Oasis | Remote | Mar 2025 – Oct 2025 (8 Months)

- Delivered real-time action recognition systems achieving 92%+ accuracy in multi-person pose tracking and gesture classification across diverse environments, using ST-GNN, LSTM, Transformers, and MediaPipe.
- Shipped end-to-end agentic AI pipelines by integrating n8n workflow automation with Large Language Models (LLMs) for memory management, intent classification, and autonomous task execution, reducing manual intervention by 70%.
- Deployed facial authentication and identity verification pipelines for secure access control, processing 500+ verifications daily at 98% accuracy using deep learning-based facial recognition models.
- Designed anomaly detection systems using autoencoders and temporal embeddings to flag abnormal movements in surveillance and fitness datasets, cutting false-positive rates by 40%.

PROJECTS

Chronicle – Autonomous Research Agent Orchestration | LINK **Jan – Feb. 2026**

- Built an autonomous AI research agent that transforms a single research goal into deeply-analyzed, multi-format reports by executing a self-correcting 8-phase pipeline of 75+ targeted searches using Google Gemini API and real-time web grounding.

NeuroInsight AI – 3D Brain Tumor Segmentation & ROI Detection | LINK **Aug – Nov 2025**

- Designed a 3D U-Net deep learning model using TensorFlow and Python for accurate brain tumor segmentation and ROI detection in volumetric MRI scans, achieving high Dice Similarity Coefficient scores on the BraTS benchmark dataset.

SymptomIQ – AI-Powered Health Condition Predictor | LINK **Jan – Feb 2025**

- Built a Flask web application using TensorFlow and Google Gemini API that predicts health conditions from user symptoms with 90%+ accuracy and provides personalized doctor recommendations via geolocation integration.

DeepFake Detector – Image Authenticity Classifier | LINK **Jan – Feb 2025**

- Developed a CNN-based deepfake detection tool using TensorFlow, OpenCV, and Flask that classifies uploaded images as FAKE or REAL with 95%+ accuracy across 10,000+ real and manipulated face images.

Fake News Detection System – NLP Project | LINK **Dec 2024**

- Built an NLP-based fake news classifier using NLTK, TF-IDF vectorization, and Scikit-Learn achieving high F1-scores on a 20,000+ article dataset through automated text preprocessing and classification.

Automated Power Monitoring System – IoT & Computer Vision | LINK **Sep – Nov 2024**

- Developed a real-time human presence detection system using OpenCV and IoT to automate appliance control and optimize energy consumption, winning the Intra-Institutional Startup Competition.

ACHIEVEMENTS

- Won 1st Place in the Intra-Institutional Startup Competition, selected among 50+ competing teams for an AI-driven energy optimization system.
- Competed in 3 national-level 24-hour hackathons — Kalinga Hackathon, Hackomania, and HackGenX — building AI/ML solutions under time-constrained environments.
- Earned the Google Cloud Certified Generative AI Leader (GAIL) certification.
- Earned advanced certifications in Python, C, and SQL from HackerRank, and an AI & Machine Learning certification from Teach Nook covering neural networks and deep learning.